

Acoustic Data-based Characterization of Incipient Boiling for Space Applications

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Abstract—Understanding and detecting early-stage boiling in cryogenic fuel tanks is critical for ensuring safety in space missions. Traditional thermal sensors used by NASA often lack the sensitivity to capture the onset of boiling events, particularly under microgravity, where fluid dynamics differ significantly from Earth. This work presents a machine learning framework that uses acoustic emissions to classify boiling behaviors and identify incipient boiling phases before pressure fluctuations compromise cryogenic tank stability. Drawing from 441 experiments conducted by NASA, we extracted time and frequency-domain features from high-resolution accelerometer data. Using these features, we developed an unsupervised clustering pipeline to discover latent boiling regimes, and a supervised decision tree classifier to predict them using interpretable signal features. A custom web application was created to enable NASA researchers to either explore and interpret the current trained models or upload new runs for classification. This tool provides real-time predictions and embeds new data points within the cluster map, effectively combining both supervised and unsupervised analysis. Our approach offers a promising enhancement to existing cryogenic fuel monitoring systems, enabling faster, non-intrusive, and more granular detection of mission-critical boiling dynamics in space.

Index Terms—Boiling regimes, Acoustic data, Cryogenic fuel tanks, Machine learning, Unsupervised learning, Supervised learning, K-Means Clustering, Decision trees

I. INTRODUCTION

Cryogenic fuels are essential to modern spaceflight, providing high energy density and efficient propulsion. Storing these fuels safely is a complex challenge, particularly under microgravity conditions, where conventional convective cooling is reduced and localized heating can result in unstable boiling. In such environments, even minor heat leaks along the tank walls can superheat the fluid, initiating incipient boiling that rapidly generates vapor. This sudden transition can trigger pressure spikes severe enough to rupture the tank, endangering missions and equipment.

NASA’s current approach to monitoring cryogenic tank stability relies heavily on thermal sensors. While effective at capturing temperature trends, these sensors are not well suited for detecting the earliest acoustic

signatures of boiling, especially in dynamic microgravity environments. These sensors often provide delayed feedback, creating a possibility of pressure escalation without warning.

We propose that boiling events produce distinct acoustic signatures, detectable via accelerometers mounted on the tank exterior. The formation, growth, and collapse of vapor bubbles each emit energy that propagates through the tank walls and can be captured as sharp spikes in acoustic data. These patterns vary based on boiling regime, offering a promising nonintrusive signal for early detection.

In this work, we build on NASA’s vision for acoustic-based monitoring by combining signal processing with machine learning. We apply unsupervised clustering to uncover latent boiling regimes, and train a supervised decision tree to predict them based on interpretable signal features. A custom-built Dash web application supports both retrospective analysis and prospective classification, allowing researchers to explore existing model structures or upload new CSV data to classify a run and inspect its cluster positioning in real time.

This system facilitates rapid insight generation, interpretable modeling, and spatial analysis of boiling dynamics, aiming to enhance the safety and intelligence of cryogenic fuel monitoring systems.

II. DATASET DESCRIPTION

We analyze data from 441 boiling experiments conducted by NASA, each ranging from 1 to 30 seconds in duration.

A. Experimental Setup

Each experiment consists of a spherical tank filled with liquid, equipped with accelerometers mounted on the tank walls, as shown in Figure 1. During each run, a controlled heat source induces boiling near the tank wall, and the acoustic emissions are captured by the accelerometers. Two sensors record the signal at a sampling rate of 10,000 Hz, but we focus our analysis on a single channel due to signal redundancy.

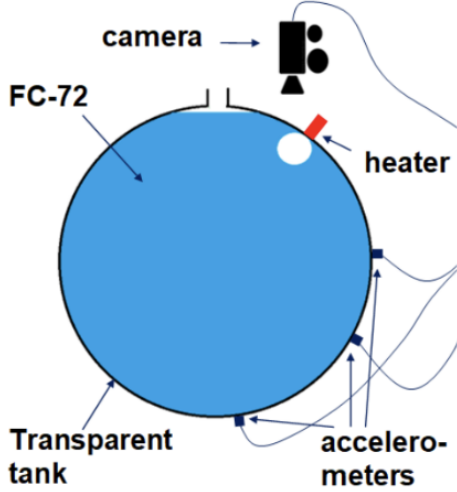


Fig. 1: Experimental setup: cryogenic tank with accelerometers.

B. Data Format and Visualization

Each run is stored in a CSV file containing time-stamped accelerometer values. We visualize this raw data in the time domain as amplitude versus time in Figure 2, revealing distinct peaks that correspond to bubble nucleation events.

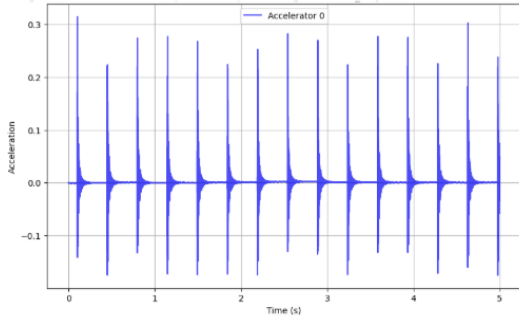


Fig. 2: Raw time-domain signal from a single experimental run.

C. Physical Interpretation

Both the time-domain and spectral representations emphasize the acoustic signals produced during bubble nucleation and collapse in a boiling liquid, especially under conditions simulating space environments. As superheated liquid reaches its boiling point near a heat source, bubbles form and grow rapidly before collapsing or detaching. This sudden formation and collapse of vapor bubbles releases energy in the form of energy waves, which manifest as sharp-spikes in the time domain signal. These spikes are linked to the behavior

of boiling and bubble activity, with different boiling regimes, such as quasi-homogeneous or heterogeneous nucleation, producing distinct acoustic signals. In microgravity conditions, like those in space, convection is suppressed, leading to more intense superheating and potentially explosive boiling, which would be particularly pronounced in acoustic signals as distinct spikes.

III. FEATURE ENGINEERING

We extract features from both time and frequency domains, giving us multiple perspectives of boiling signals from which we gain insight.

A. Time Domain Features

1) *Peak Detection*: We use SciPy's `find_peaks` with adaptive thresholds. The vertical threshold is based on the greater of the 99.5th percentile or 10% of the max value, clamped between 0.015 and 0.1. The horizontal threshold scales inversely with peak amplitude to avoid false detections.

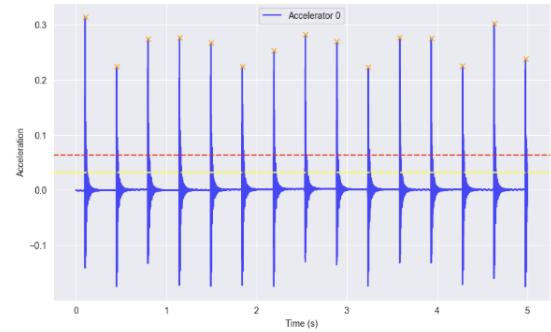


Fig. 3: Detected peaks using adaptive thresholding.

2) *Derived Peak Features*: From each run, we calculate:

- Number of statistically significant dominant rhythms
- Mean, median, and standard deviation of peak-to-peak intervals
- Peak magnitude statistics (max, median, std)
- Peaks per second
- Sum of peak magnitudes per second

The algorithm for detecting the number of statistically significant dominant rhythms is detailed in Appendix A.

3) *Additional Metrics*: We include:

- Percent of time above threshold
- Global signal standard deviation

B. Frequency Domain Features

1) *Spectral Transformation*: We apply a high-pass Butterworth filter, then use FFT and Welch's method to compute spectral features, as seen in Figure 4.

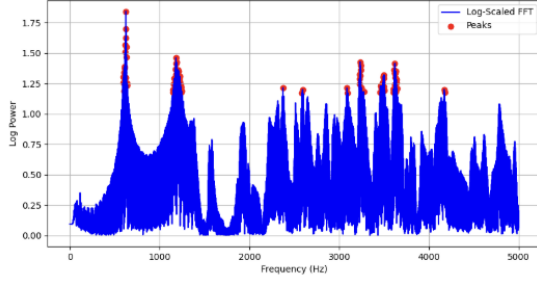


Fig. 4: FFT of filtered accelerometer signal.

2) Extracted Features:

- Dominant frequency
- Number of spectral peaks
- Weighted average frequency
- Mean and standard deviation of spectral power
- Spectral entropy

IV. UNSUPERVISED LEARNING

A. Pipeline

We implemented an unsupervised learning pipeline to analyze acoustic boiling regime data. The pipeline consists of the following steps:

- **Feature Standardization:** All extracted features are standardized using `StandardScaler` to ensure each feature contributes equally to the analysis.
- **Dimensionality Reduction:** Principal Component Analysis (PCA) is applied to the standardized features. The first 10 principal components are retained, capturing approximately 97% of the total variance in the dataset. This reduces noise and highlights the most informative patterns. (See Figure 5 for the scree plot, which shows the cumulative variance explained by varying numbers of principal components.)
- **Clustering:** KMeans clustering is performed in the reduced PCA space, enabling the identification of natural groupings and patterns within the data without relying on labels.

B. Cluster Analysis

Varying k revealed distinct groupings, interpreted using PCA loadings and rhythm patterns. Figure 6, shown on the next page, displays the data plotted in the PCA space, with k-means clustering applied with $k = 4$.

C. Cluster Validation

To assess clustering quality and interpret cluster meanings, we compute the Szymkiewicz–Simpson overlap between each cluster and true class:

$$\text{Sim}(A, B) = \frac{|A \cap B|}{\min(|A|, |B|)}$$

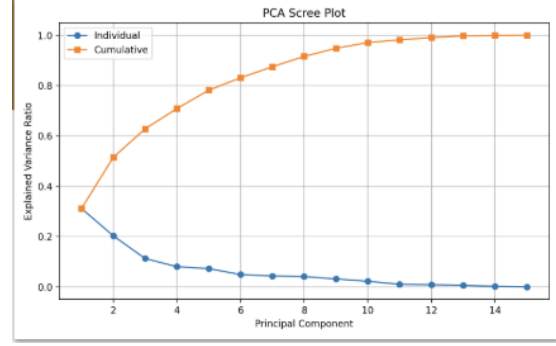


Fig. 5: Scree plot of PCA variance.

Values close to 1 indicate a strong match between a cluster and a class, whereas values near 0 suggest little to no overlap.

Figure 7 presents the resulting overlap matrix with 7 clusters.

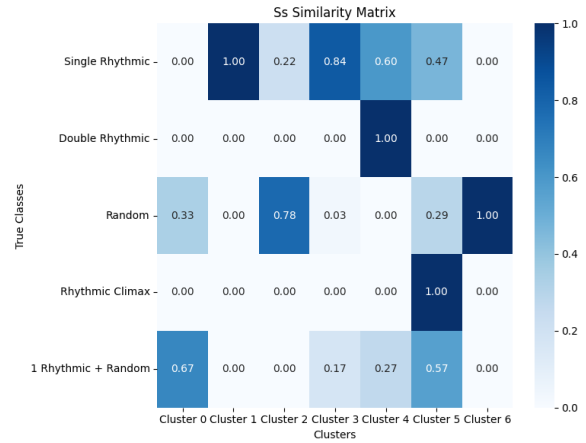


Fig. 7: Cluster validation using Szymkiewicz–Simpson overlap coefficient.

V. SUPERVISED LEARNING

A. Training and Evaluation

The supervised learning stage of our pipeline leverages labeled data to train a Decision Tree classifier for boiling regime classification. After feature extraction, categorical labels are encoded and the data is split into training and validation folds. Cross-validation is used to provide robust estimates of model performance.

Model training and evaluation are fully automated: the decision tree is fit on the training data, and predictions are generated for each fold. Performance metrics, including per-class F1-scores and a weighted F1-score, are computed to assess classification accuracy across both dominant and minority classes. The pipeline also generates visualizations such as feature importance plots

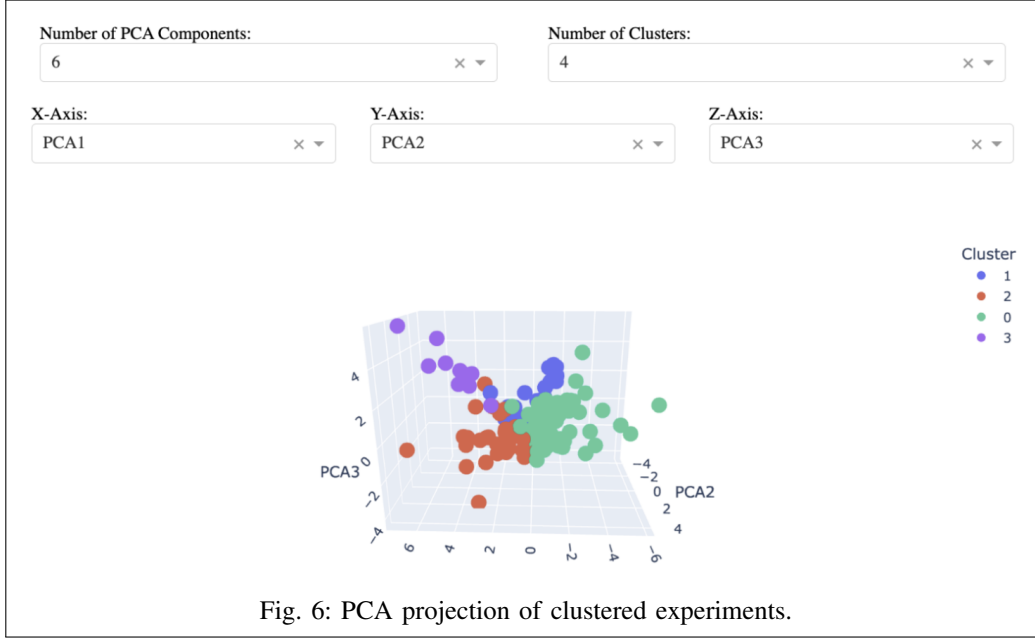


Fig. 6: PCA projection of clustered experiments.

and confusion matrices, which are saved for downstream analysis and reporting.

This approach ensures that model evaluation is consistent, transparent, and easily extensible to new data.

B. Results

- **Single Rhythmic:** F1 = 85.2%
- **Random:** F1 = 83.4%
- **Noise:** F1 = 87.4%
- Rare classes performed poorly due to low support
- Weighted F1-score = 79.1%

Figure 8 shows the relative feature importance from the decision tree.

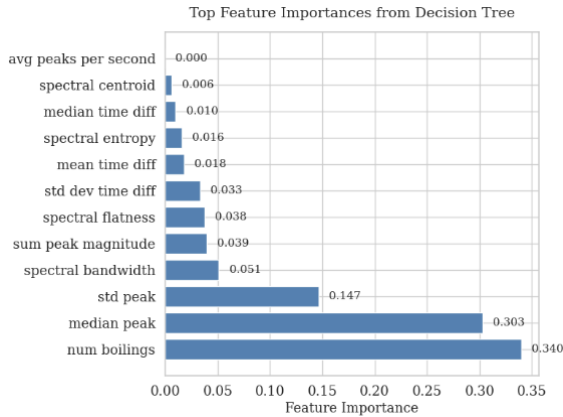


Fig. 8: Feature importance from decision tree.

Based on this, we can see that the most important features are:

- Number of boilings
- Median and standard deviation of peak magnitudes
- Spectral bandwidth

C. Comparison with Other Methods

Alternative modeling approaches, such as XGBoost and neural networks, were also explored for boiling regime classification. However, these models consistently underperformed when compared to the Decision Tree classifier. Across both dominant and minority classes, they yielded lower F1-scores and failed to identify meaningful distinctions among rhythmic patterns. Moreover, their lack of interpretability limited our ability to extract actionable insights or visualize decision boundaries—a key requirement for this domain. In contrast, the Decision Tree offered a reliable and transparent solution, aligning well with the interpretability demands and data limitations of this study.

D. Handling Class Imbalance

We evaluated several techniques to address the imbalance in class distribution, including class weighting, random oversampling, and Synthetic Minority Over-sampling Technique (SMOTE). Despite these efforts, improvements remained modest due to the severe sparsity in certain labels. Minority classes such as *Double Rhythmic* and *Rhythmic Climax* were especially affected, with insufficient samples to enable generalizable learning. These limitations suggest that future progress may depend more on collecting targeted additional data or employing domain-informed data augmentation techniques to enrich underrepresented classes.

E. Confusion Matrix Analysis

Figure 9 presents the confusion matrix aggregated across all five folds of cross-validation. Most misclassifications occurred between acoustically or structurally similar boiling regimes. Notably:

- *Single Rhythmic* was occasionally misclassified as *Rhythmic Climax* or *Rhythmic & Random*.
- *Random* sometimes overlapped with both *Noise* and *Rhythmic Climax*.
- *Double Rhythmic* and *Rhythmic Climax* classes yielded very low true positive rates, reinforcing the issue of class sparsity.

These trends suggest that while the model is effective at detecting dominant rhythmic structures, it struggles to separate closely related or infrequent patterns. This difficulty likely stems from overlapping feature representations and limited training data for the minority classes.

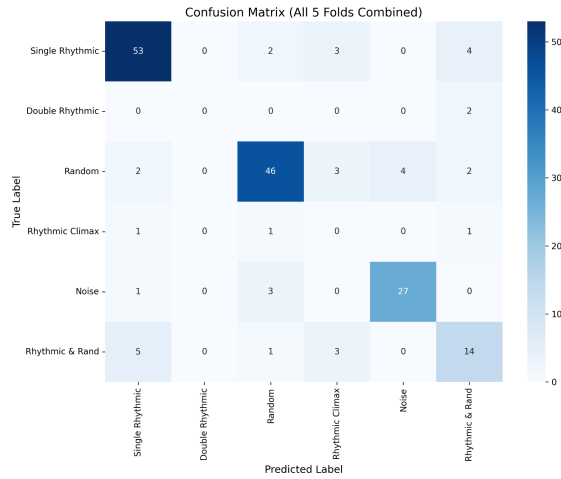


Fig. 9: Confusion matrix.

VI. WEB APPLICATION FOR INTERACTIVE EXPLORATION AND CLASSIFICATION

To bridge the gap between model development and researcher usability, we built an interactive web application using Dash, as seen in Figure 10. This web interface serves as both an analytical and interpretive tool, enabling researchers at NASA to explore clustering results, visualize decision boundaries, and classify new experimental runs with minimal technical overhead.

The application supports two primary modes of use:

- 1) **Model Exploration Mode (Figure 11):** Users can inspect the behavior of the current trained unsupervised and supervised models. This includes an interpretable decision tree visualization rendered as a zoomable, pannable SVG, allowing users to trace the logical flow of predictions and understand how specific features influence model decisions.

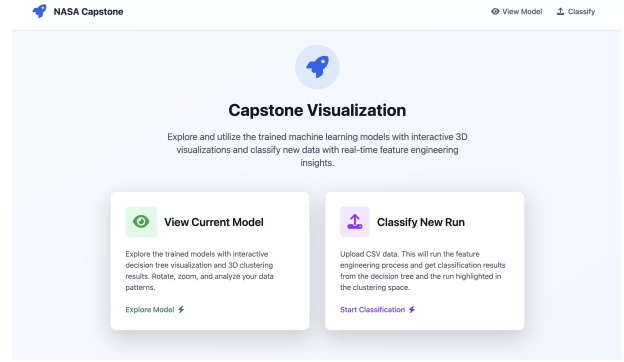


Fig. 10: Web app landing page presenting two primary modes: model exploration and new run classification.

In parallel, users can explore clustering results in a 3D PCA-reduced space. Researchers may dynamically adjust the number of clusters and select different PCA axes to reveal latent structure in the data and identify potential regime separability.

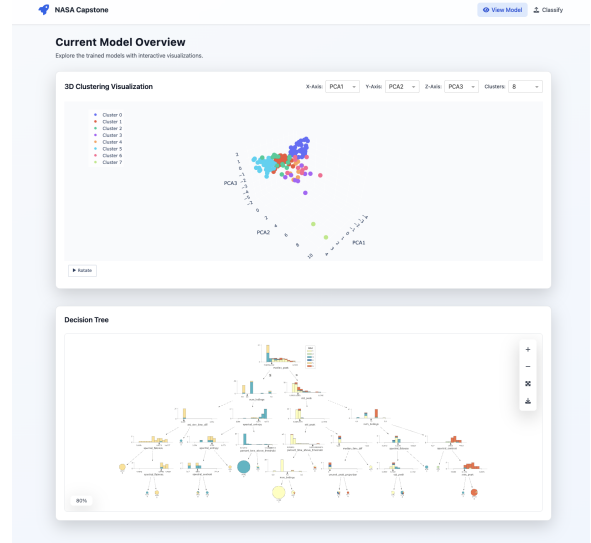


Fig. 11: Model Exploration mode.

- 2) **Classification Mode (Figure 12):** Users can upload new experimental runs in a CSV format. Once uploaded, the application automatically applies the full feature engineering pipeline, including time-domain and frequency-domain transformations. The extracted features are passed through the trained decision tree classifier, which outputs a predicted boiling regime, class probabilities, and a list of influential features. Simultaneously, the new run is projected into the PCA space and overlaid on the existing cluster visualization. This dual approach allows researchers to assess classification

outcomes in both a supervised context (via the decision tree) and an unsupervised context (via clustering), offering a more holistic understanding of the run's regime.

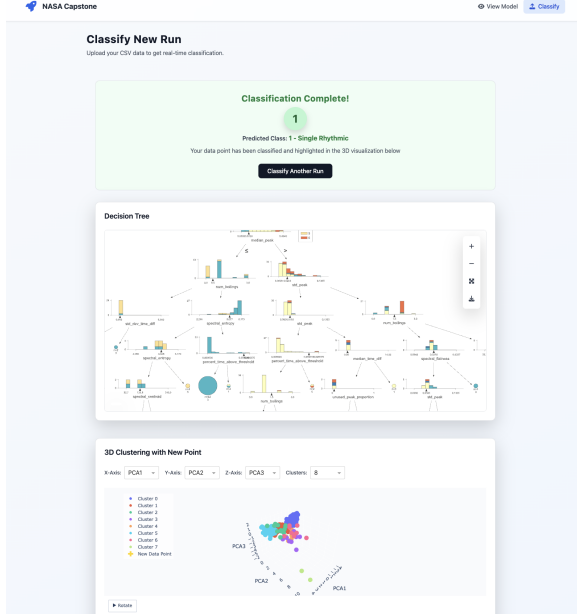


Fig. 12: Classification mode output: predicted boiling regime and projection of the new data point in PCA cluster space.

Together, these capabilities transform our machine learning pipeline into a robust, real-time decision support tool. The web application enables researchers to interactively explore both the unsupervised (clustering) and supervised (decision tree) models, validate predictions, and understand key signal characteristics driving boiling regime classification. Additionally, the modular architecture of our codebase supports future extensions and modifications, whether to the regimes, experimental runs, decision tree logic, or clustering pipeline. This flexibility enables potential integrations such as onboard deployment with spacecraft sensor streams or adaptive model retraining as new data becomes available.

VII. CONCLUSION

We present a machine learning pipeline for detecting early-stage boiling in cryogenic fuel systems using high-resolution acoustic emissions. Our approach leverages both time- and frequency-domain signal processing to extract more responsive boiling indicators. These features feed into a hybrid modeling framework consisting of unsupervised clustering for regime discovery and supervised decision tree classification for real-time interpretability and classification.

To support deployment and usability in practical research settings, we developed a custom web application that allows researchers to visualize model behavior, inspect boiling regime boundaries, and classify new experimental runs. This interface bridges the gap and makes advanced analytics accessible and actionable.

Our results demonstrate that acoustic-based detection, when paired with machine learning, offers a viable path forward for enhancing safety and situational awareness in cryogenic systems. The modular and extensible design of our pipeline also enables ongoing adaptation, supporting future extensions such as retraining with new data, or expansion to other discovered regimes. Ultimately, this work lays the foundation for building safer, smarter, and more autonomous cryogenic fuel monitoring systems for future space missions.

APPENDIX A DETECTION OF DOMINANT RHYTHMIC PATTERNS ALGORITHM

Purpose

This algorithm aims to detect recurring rhythmic patterns in a sequence of detected acoustic peaks. It evaluates every possible candidate interval derived from peak differences, tests each rhythm statistically, and consolidates similar patterns while removing weak or redundant ones. The result is a clean list of dominant intervals that likely represent true underlying rhythmic structure in the data.

Plain English Overview of Algorithm

0) Estimate Noise Level

Before any rhythm analysis, the algorithm estimates the natural variability in peak timing and amplitude. This standard deviation is used to compute a margin of tolerance when evaluating rhythmic alignment and to calibrate statistical tests later in the process.

1) Generate Rhythm Candidates

Every pair of peaks is used to define a potential rhythm interval $d = x_j - x_i$, anchored at x_i . Only differences that allow a sufficient number of repetitions (typically based on $\sqrt{n}$) within the signal duration are retained.

2) Count Hits Per Candidate

For each candidate, the algorithm walks forward in time from the anchor using $t_k = \text{anchor} + k \cdot d$, checking whether predicted steps align closely with real peaks, both in the x and y direction. Each match is a “hit,” and both the number of hits and total prediction attempts (tries) are recorded.

3) Prune Candidates via Binomial Test

The algorithm tests each candidate using a one-sided binomial hypothesis test. It asks: “Is the

observed hit rate significantly better than random chance?” Candidates that fail this test (after adjusting for multiple comparisons) are removed.

4) **Group Similar Candidates**

Candidates that share a large number of overlapping hits are grouped using a similarity metric (Szymkiewicz–Simpson coefficient). Within each group, the strongest candidate is retained and others are marked as absorbed.

5) **Final Filtering by Support**

As a final step, candidates that don’t have enough total support (from both their own hits and absorbed ones) are removed. This ensures the algorithm outputs only the most robust rhythmic patterns.

The pseudocode for the algorithm is presented in Algorithm 1.

Example Walkthrough Using Peak Time Array

We demonstrate the full algorithm using the following observed peak times from an acoustic signal:

$$\text{peaks} = \left\{ \begin{array}{cccc} 0.1035, & 0.3391, & 0.5746, & 0.8100, \\ 1.0454, & 1.2791, & 1.5160, & 1.7496, \\ 1.9866, & 2.2220, & 2.4574, & 2.6930, \\ 2.9286, & 3.1641, & 3.3999, & 3.6359, \\ 3.8722, & 4.1088, & 4.3458, & 4.5828, \\ 4.8197 \end{array} \right\}$$

The following plot shows the original acoustic signal, with peaks marked:

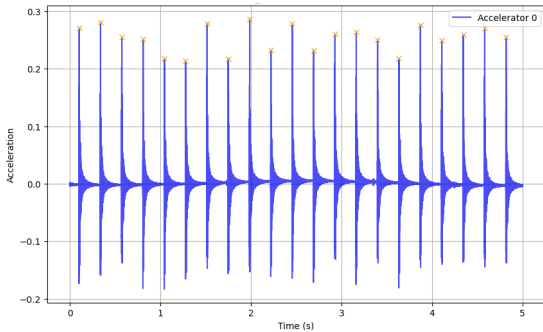


Fig. 13: Acoustic signal with extracted peaks.

Step 0: Estimate Noise Level: We calculate the global standard deviation of peak-to-peak spacing:

$$\text{SD} = 0.00091 \Rightarrow \text{Margin} = z \cdot \text{SD} = 0.0015$$

The probability of a random hit in a margin window is estimated as:

$$p_{\text{null}} = \frac{2 \cdot \text{margin} \cdot \# \text{peaks}}{\text{run length}} = \frac{2 \cdot 0.0015 \cdot 21}{5} \approx 0.0126$$

Algorithm 1 Detection of Dominant Rhythmic Patterns in Acoustic Peaks

- 1: **Input:** Sorted peak times $x = [x_1, x_2, \dots, x_n]$, amplitudes $y = [y_1, y_2, \dots, y_n]$, total duration T
- 2: **Output:** Filtered list of dominant rhythmic candidates

Step 0: Estimate Noise Level

- 3: $\sigma_x \leftarrow \text{estimate_std}(x)$ $\triangleright$ Timing variability
- 4: $\sigma_y \leftarrow \text{estimate_std}(y)$ $\triangleright$ Amplitude variability
- 5: $m_x \leftarrow z_\alpha \cdot \sigma_x$, $m_y \leftarrow z_\alpha \cdot \sigma_y$
- 6: $p_{\text{null}} \leftarrow \min(1, \frac{2 \cdot m_x \cdot n}{T})$

Step 1: Generate Rhythm Candidates

- 7: **for** $i = 1$ to $n - 1$ **do**
- 8: **for** $j = i + 1$ to n **do**
- 9: $d \leftarrow x_j - x_i$
- 10: **if** $x_j + \sqrt{n} \cdot d > T + m_x$ **then**
- 11: **break**
- 12: Add candidate (d, x_i) to candidate list

Step 2: Count Hits Per Candidate

- 13: **for** each candidate (d, a) **do**
- 14: Walk forward from anchor a in steps of d
- 15: At $t_k = a + k \cdot d$, estimate amplitude $\bar{y}_k$
- 16: Count a hit if $\exists x_i \in [t_k \pm m_x]$ and $y_i \in [\bar{y}_k \pm m_y]$
- 17: Record number of hits and tries

Step 3: Prune via Binomial Test

- 18: **for** each candidate with h hits and t tries **do**
- 19: $\alpha' \leftarrow \alpha/R$ $\triangleright$ Bonferroni correction
- 20: $p \leftarrow \text{BinomialTest}(h - 2, t - 2, p_{\text{null}})$
- 21: **if** $h < \sqrt{n}$ or $p \geq \alpha'$ **then**
- 22: Remove candidate

Step 4: Group Similar Candidates

- 23: Construct similarity graph with edge if
- 24: $\text{Szymkiewicz-Simpson}(A_i, A_j) \geq \tau$
- 25: **for** each connected component **do**
- 26: Keep candidate with highest centrality
- 27: Absorb unique hits from others into dominant

Step 5: Final Filtering by Support

- 28: **for** each remaining candidate **do**
- 29: **if** (hits + absorbed hits) $< \sqrt{n}$ **then**
- 30: Remove candidate
- 31: **return** Remaining candidates as dominant rhythmic intervals

Step 1: Generate Rhythm Candidates: For each peak pair (i, j) , we compute the difference $d = x_j - x_i$ and retain it as a candidate if it allows at least $\sqrt{n}$ steps within the duration.

Example early candidates:

- Candidate 0: $d = 0.2356$, anchor = 0.1035
- Candidate 1: $d = 0.4711$, anchor = 0.1035
- Candidate 2: $d = 0.7065$, anchor = 0.1035

This process generates a total of 41 candidates.

Candidate ID	d (sec)	Anchor	Hits	Tries
0	0.2356	0.1035	20	20
1	0.4711	0.1035	10	10
2	0.7065	0.1035	6	6
3	0.9429	0.1035	5	5
4	1.1785	0.1035	3	4

Step 3: Prune Candidates via Binomial Test: Each candidate is evaluated using a one-sided binomial test under:

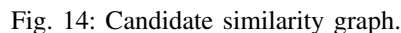
A Bonferroni adjustment is applied using α/R for R candidates. Clusters failing the test are removed.

$$\text{Sim}(A, B) = \frac{|A \cap B|}{\min(|A|, |B|)}$$

Step 5: Final Filtering by Support: Candidates with too few total hits (direct + absorbed) are removed. In this example, only rhythms with $\geq \sqrt{21} \approx 5$ hits are retained.

Detected Rhythm: ID = 0, $\mu = 0.2356$ seconds,
aligned with 20 out of 20 peaks

The figure below shows the acoustic signal with peaks labeled according to their assigned rhythmic candidate. Each candidate is represented by a distinct color—if



Example on Double Rhythmic Signal

Notes

- Peaks that align with a candidate's predicted steps—but were not part of its original cluster—are marked as absorbed, as seen in Figure 16.
- Absorbed peaks are visualized using a lighter shade of the candidate's color to distinguish them from originally clustered peaks.
- Peaks not matched to any candidate rhythm are considered unused and are displayed in gray and likely correspond to random boiling. A higher proportion of unused peaks may indicate random boilings.

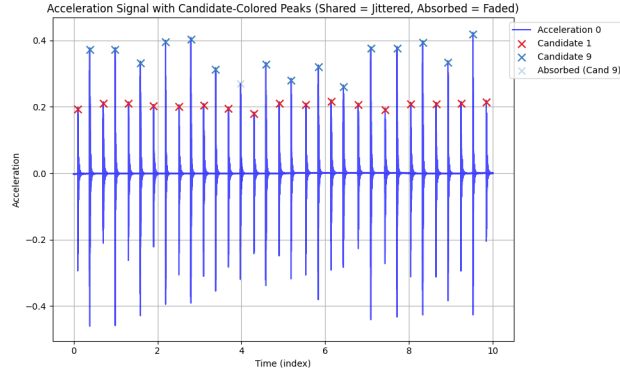


Fig. 16: Labeled double rhythmic acoustic signal with candidate assignments.

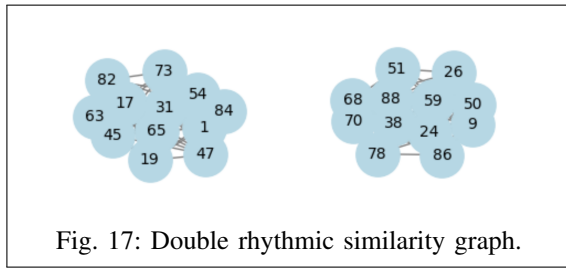


Fig. 17: Double rhythmic similarity graph.

APPENDIX B

CLUSTERING INTERPRETATION AND PRINCIPAL COMPONENTS

A. Overview

This appendix presents a sample clustering outcome from our web application, using three principal components (PC1, PC2, PC3) and grouping the data into three clusters. The figure below visualizes each experimental run in 3D space, with colors indicating cluster assignments.

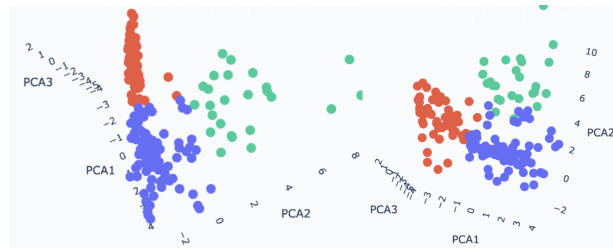


Fig. 18: 3D projection of boiling experiments clustered by PCA components.

B. PCA Loadings and Cluster Interpretation

Principal Component Analysis (PCA) was used to reduce the complex, multi-dimensional feature space into a few principal axes that capture the dominant patterns in the boiling regime data. By examining the contributions

of each original feature to the principal components, we can interpret the underlying physical and statistical characteristics that differentiate the data.

Principal Component 1 (PCA1):

The first principal component primarily reflects the overall intensity and activity of the boiling signal. High values along this axis correspond to runs with more frequent and larger acoustic events, as well as signals that are more complex and less uniform. In contrast, lower values indicate quieter runs with fewer, smaller, or more uniform events, and a more regular or “flat” signal profile.

Principal Component 2 (PCA2):

The second principal component is most influenced by the temporal structure of the boiling events and the presence of unused or less prominent acoustic features. High values along this axis are associated with runs where the timing between events is more variable or extended, and where a greater proportion of potential events are not strongly expressed in the signal. This suggests a regime with more irregular or sporadic boiling activity.

Principal Component 3 (PCA3):

The third principal component captures variation related to the complexity and distribution of energy within the signal. High values along this axis are associated with runs where the acoustic energy is more broadly distributed across different frequencies and where the signal exhibits greater unpredictability or irregularity. This suggests that PCA3 distinguishes between boiling regimes with more complex, less periodic acoustic patterns and those with simpler, more predictable structures.

Cluster Interpretation in PCA Space:

- **Cluster 0 (green):** This group is distinguished by higher values along the second principal component, indicating runs with more irregular timing between events and a greater presence of subtle or less pronounced acoustic activity.
- **Cluster 1 (blue):** This cluster stands out for its higher values along the first principal component, representing runs with more intense, frequent, and complex boiling activity.
- **Cluster 2 (red):** This group is characterized by lower values along the first principal component, corresponding to runs with less intense, more uniform, and quieter boiling signals.

This analysis shows that the main axes of variation in the data correspond to differences in both the intensity and regularity of boiling activity, as well as the presence of subtle or sporadic events. The clustering in PCA space thus provides a physically meaningful separation of boiling regimes, reflecting both the strength and the temporal structure of the acoustic signals.

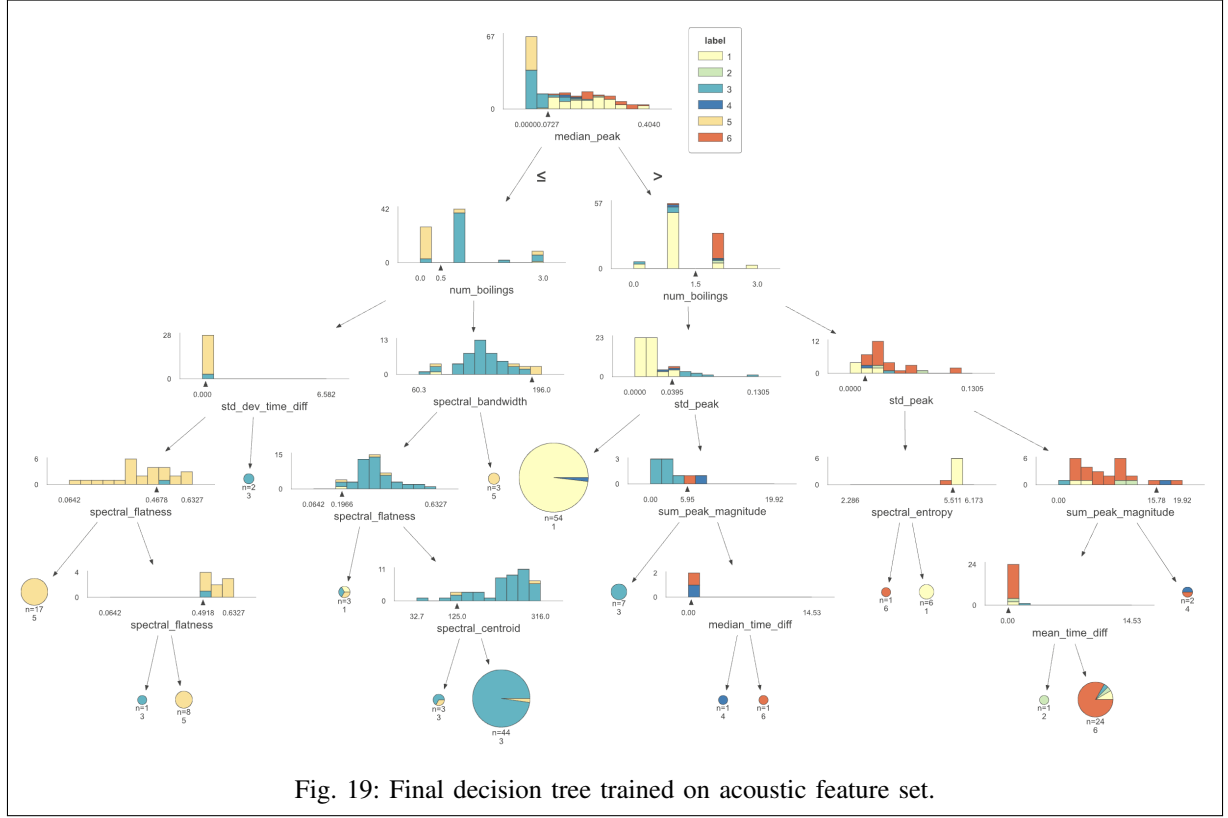


Fig. 19: Final decision tree trained on acoustic feature set.

APPENDIX C

DECISION TREE SUMMARY AND METRICS

A. Trained Decision Tree

Figure 19 shows our final trained decision tree. This model was constrained to a maximum depth of five to improve interpretability and prevent overfitting.

TABLE I: Decision Tree Classification Report

Class	Support	Precision	Recall	F1-Score
Single Rhythmic	62	86.56	85.48	85.22
Double Rhythmic	2	0.00	0.00	0.00
Random	57	87.54	80.70	83.36
Rhythmic Climax	3	0.00	0.00	0.00
Noise	31	90.16	87.09	87.38
Rhythmic & Rand	23	68.98	60.86	58.11
Weighted Avg	178	82.80	78.65	79.11

B. Interpretation

The decision tree performed well for high-support classes, achieving F1-scores above 83% for Single Rhythmic, Random, and Noise. However, the model struggled with underrepresented classes (Double Rhythmic, Rhythmic Climax), resulting in F1-scores of zero due to insufficient training data. These results underscore the need for class-balancing techniques and possibly more data or ensemble methods to boost performance on rare classes.

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